

Linear Expansion

From a real bridge to a law: why bridges need a joint

Physics · 3A · 18 September 2026

Where we left off

(slide 2)

Before we start, a recap of what we did on 11 September, in three questions.

Before moving on. What is the difference between temperature and heat? What does it mean for two objects to be in thermal equilibrium? Why can temperature never be negative in kelvin?

Temperature is a property of an object; heat is energy that passes from one object to another when they are at different temperatures. Two objects are in thermal equilibrium when there is no net transfer of energy between them: they are at the same temperature. In kelvin, zero is the lowest possible temperature, the one of minimum thermal agitation, and you cannot go below it.

A real bridge

(slide 3)

The Golden Gate Bridge in San Francisco. The main span is 1 275 m of steel long. Over the year the temperature goes from -15°C to 40°C .

Before moving on. By how much does the span expand between the coldest and the hottest day: a few millimetres, a few centimetres or almost a metre? Write down your answer: we will come back to it at the end of the chapter.

The bridge is real and the problem is a standard exercise from *University Physics* (OpenStax). To answer it we need a law.

The law of linear expansion

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If I heat a solid, its length increases (figure 1). The change in length is proportional to the initial length and to the change in temperature:

$$\Delta L = \alpha L_0 \Delta T$$

ΔL is the change in length (m); α is the **linear expansion coefficient**, which depends on the material and is expressed per degree, in $1/^{\circ}\text{C}$; L_0 is the initial length (m); $\Delta T = T - T_0$ is the change in temperature. ΔT has the same value in $^{\circ}\text{C}$ and in K.

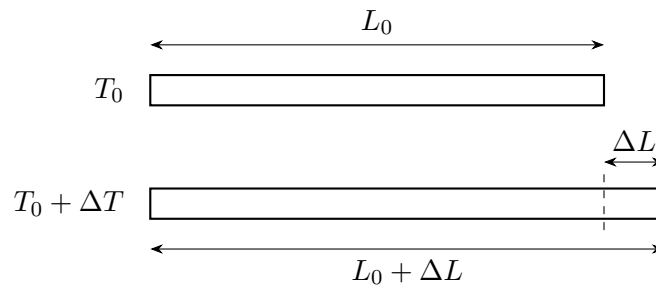


Figure 1: A bar before and after heating. The change ΔL is greatly exaggerated: on a real bar it is too small to see (schematic, not to scale).

Playing with the law

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Before moving on. If I double L_0 , what happens to ΔL ? And if ΔT changes sign?

If L_0 doubles, ΔL doubles: they are proportional. If ΔT is negative, so is ΔL : the solid contracts, with the same law. In the classroom simulation the bar is enlarged 200 times, because on a real bar the change is too small to see.

The coefficients

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For the calculations we need α for a few materials.

Material	$\alpha / 10^{-6}/^{\circ}\text{C}$
Steel, iron	12
Copper	17
Brass	19
Aluminium	23
Glass	9

Typical textbook values near room temperature.

Aluminium expands about 2.5 times more than glass, for the same length and the same ΔT .

A worked example

(slide 7)

An iron rail 20 m long goes from -10°C to 35°C . How much does it expand?

We set it up first: data, unknown, principle, then the calculation.

1. $\Delta T = 35 - (-10) = 45^{\circ}\text{C}$. First the change in temperature.
2. $\Delta L = 12 \times 10^{-6} \times 20 \times 45$: apply the law.
3. $\Delta L = 0.0108 \text{ m} = 1.08 \text{ cm}$: result with consistent units.

When steel can't expand

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The rail has a worrying side. Modern rails are welded into one continuous ribbon: no more of the old joints with a gap. In summer the steel heats up far more than the air and wants to expand, but the sleepers hold it in place: huge stress builds up. If the stress exceeds the limit, the track bends sideways. It is called a *sun kink*.

In July 2002 an Amtrak train derailed near Washington, in Maryland, injuring over 100 people. According to reports at the time, the derailment was linked to a sun kink.

The law $\Delta L = \alpha L_0 \Delta T$ explains why: if ΔL cannot happen, the stress goes up instead.

Source: AP and CNN reports on the July 2002 derailment at Kensington, Maryland; engineering literature on sun kinks.

Now it's your turn

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For each exercise, set up first: data, unknown, principle, justification, equation. Then calculate.

1. A copper cable 50 m long goes from 5 °C to 40 °C. By how much does it expand?
2. A steel rail 15 m long goes from −5 °C to 40 °C. By how much does it expand? (It is the rail of the sun kink.)
3. A copper bar 3 m long goes from 20 °C to 150 °C. By how much does it expand?
4. An aluminium bar 2.000 m long at 20 °C expands by 3.68 mm when heated to 100 °C. Find α and check that it matches the value in the table.

The solutions are at the end of the chapter.

When the simple model isn't enough

(slide 10)

The Eiffel Tower: 300 m of puddled iron. If it were a single bar, with $\Delta T = 60$ °C the law predicts

$$\Delta L = 12 \times 10^{-6} \times 300 \times 60 = 0.216 \text{ m} = 21.6 \text{ cm}$$

of seasonal change. Engineers instead measure 12 cm–15 cm (figure 2).

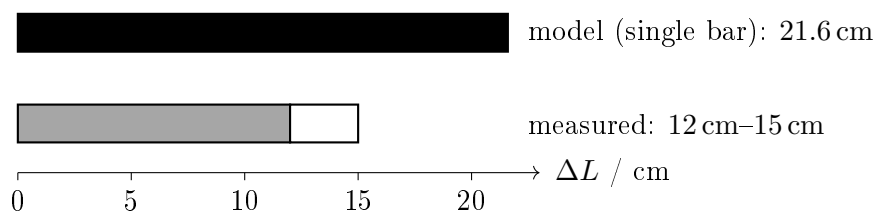


Figure 2: Seasonal change in height of the Eiffel Tower: prediction of the single-bar model and measured range.

Before moving on. Why is the measurement lower than the prediction?

Because the tower is not a bar: it is a lattice of about 18 000 pieces, not all vertical, and the sun heats one side more than the others. The law is still correct; it is the single-bar assumption that does not hold for a complex structure. Every model has a domain of validity.

Source: engineering monitoring of the Eiffel Tower, reported by The Conversation and Interesting Engineering (2025).

Back to the bridge

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Golden Gate Bridge: 1 275 m of steel, $\Delta T = 40 - (-15) = 55^\circ\text{C}$. Same law, real numbers:

$$\Delta L = 12 \times 10^{-6} \times 1275 \times 55 = 0.8415 \text{ m} \approx 0.84 \text{ m}.$$

Almost a metre. Look again at the answer you wrote down at the start.

The movement does not disappear: it is spread across the many expansion joints along the bridge. This is why bridges have joints.

From length to area and volume

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A solid expands in all directions, not just in length. For area, a side that goes from L_0 to $L_0(1 + \alpha\Delta T)$ takes the area $A = L^2$ to

$$A = A_0 (1 + \alpha\Delta T)^2 = A_0 (1 + 2\alpha\Delta T + \alpha^2\Delta T^2).$$

But $\alpha\Delta T$ is about 10^{-3} , so $(\alpha\Delta T)^2 \sim 10^{-6}$ is negligible; the same holds for the cube, in the case of volume. We get

$$\beta \approx 2\alpha, \quad \gamma \approx 3\alpha$$

$$\Delta A = \beta A_0 \Delta T, \quad \Delta V = \gamma V_0 \Delta T$$

β is the coefficient of area (superficial) expansion, γ the coefficient of volumetric expansion.

Does the hole get smaller or bigger?

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Before moving on. A metal plate has a hole. If I heat it, does the hole get smaller or bigger?

Bigger (figure 3). The hole expands as if it were made of the same material as the plate. One way to think about it: a hole is a piece of plate that is missing; if you enlarged a photograph of the plate, you would enlarge the hole too.

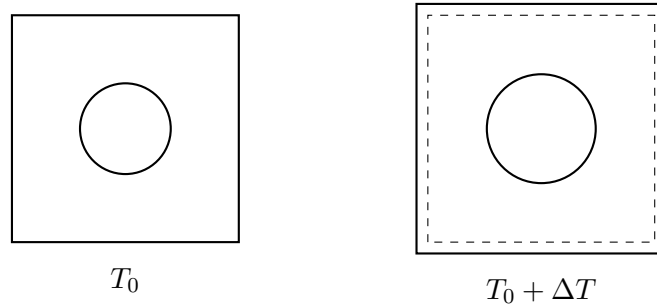


Figure 3: A plate with a hole, before and after heating. Dashed lines show the original plate and hole; the change is greatly exaggerated (schematic, not to scale).

Shrink fitting

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The same phenomenon is used on purpose. Heat a ring or a bearing: its hole expands. Slide it onto the shaft. Let it cool: it shrinks and grips tightly. It is a real technique (*shrink fitting*) used to mount wheels, gears and bearings.

Before moving on. Why not simply press the ring onto the shaft?

Because a tight fit would need a huge force and would scratch the surfaces.

Area expansion: a worked example

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A steel plate has a hole of area 5.00 cm^2 at 20°C . Heated to 220°C , how big is the hole?

1. $\Delta T = 200^\circ\text{C}$, $\beta = 2 \times 12 \times 10^{-6} = 24 \times 10^{-6}/^\circ\text{C}$: double α , don't start from scratch.
2. $\Delta A = 24 \times 10^{-6} \times 5.00 \times 200 = 0.024 \text{ cm}^2$: the area of the hole increases.
3. $A = 5.00 + 0.024 = 5.024 \text{ cm}^2$: the hole gets bigger, consistent with the fact we just saw.

The expansion tank

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Why does a car have an expansion tank? The engine coolant heats up a lot and increases in volume. Without a tank to absorb the increase, the pressure would burst the pipes. For liquids γ is generally much larger than 3α of solids.

The same physics on a planetary scale: the ocean warms and increases in volume. According to NASA, about a third of the sea level rise observed by satellites since 2004 comes from this thermal expansion, not from melting ice. The share varies from year to year: in 2024 it was about two thirds.

Source: NASA Sea Level Change, *Thermal expansion* (sealevel.nasa.gov).

Volume expansion: a worked example

(slide 18)

A brass pipe fitting has a volume of 500 cm^3 at 20°C and is heated to 80°C . By how much does its volume increase?

1. $\Delta T = 60^\circ\text{C}$, $\gamma = 3 \times 19 \times 10^{-6} = 57 \times 10^{-6}/^\circ\text{C}$: triple α .
2. $\Delta V = 57 \times 10^{-6} \times 500 \times 60 = 1.71\text{ cm}^3$: small, but measurable.

It is the same phenomenon as the expansion tank, on a different scale.

Three points to keep

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1. $\Delta L = \alpha L_0 \Delta T$: expansion depends on the material, the initial length and the change in temperature.
2. The Golden Gate Bridge expands by 84 cm: that is why bridges need expansion joints.
3. Area and volume expansion follow the same law, with $\beta \approx 2\alpha$ and $\gamma \approx 3\alpha$: from the hole that gets bigger to the expansion tank.

Next lesson, 22 September: Arduino Lab 1, where measurement error becomes central again.

Solutions to the exercises

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1. Data: $L_0 = 50 \text{ m}$, $\alpha = 17 \times 10^{-6}/^\circ\text{C}$. $\Delta T = 40 - 5 = 35^\circ\text{C}$. $\Delta L = 17 \times 10^{-6} \times 50 \times 35 = 0.02975 \text{ m} \approx 2.98 \text{ cm}$.
2. Data: $L_0 = 15 \text{ m}$, $\alpha = 12 \times 10^{-6}/^\circ\text{C}$. $\Delta T = 40 - (-5) = 45^\circ\text{C}$. $\Delta L = 12 \times 10^{-6} \times 15 \times 45 = 0.0081 \text{ m} = 0.81 \text{ cm} = 8.1 \text{ mm}$.
3. Data: $L_0 = 3 \text{ m}$, $\alpha = 17 \times 10^{-6}/^\circ\text{C}$. $\Delta T = 150 - 20 = 130^\circ\text{C}$. $\Delta L = 17 \times 10^{-6} \times 3 \times 130 = 0.00663 \text{ m} \approx 6.63 \text{ mm}$.
4. Unknown: α . From the law, $\alpha = \frac{\Delta L}{L_0 \Delta T}$. First convert mm to metres: $\Delta L = 3.68 \text{ mm} = 0.00368 \text{ m}$. $\Delta T = 100 - 20 = 80^\circ\text{C}$. $L_0 \Delta T = 2.000 \times 80 = 160 \text{ m}^\circ\text{C}$. $\alpha = \frac{0.00368}{160} = 23 \times 10^{-6}/^\circ\text{C}$: this is the value for aluminium in the table.